

HARDIK SRIVASTAVA

hardiksrwork@gmail.com | +1 (206) 773-9806 | Seattle, WA
[Portfolio](#) | [LinkedIn](#) | [LeetCode](#) | [ResearchGate](#) | [Google Scholar](#) | [GitHub](#)

EDUCATION

University of Washington <i>Master of Science in Data Science</i>	Seattle, WA Sept 2025 - Mar 2027
SRM Institute of Science and Technology <i>Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4)</i>	Chennai, TN May 2019 - Jun 2023

WORK EXPERIENCE

Graduate Research Assistant (P.I. - Prof. Zaid Harchaoui) Paul G. Allen School of Computer Science, University of Washington	Sept 2025 - Present Seattle, WA
- Architected an authorship obfuscation system by fine-tuning 16+ modular LoRA adapters (Qwen2.5-32B); built metric-driven agents to guide controllable style transfer via learned style embeddings and constrained decoding, achieving 40% privacy improvement. - Developed a gradient-free few-shot optimizer using in-context learning to dynamically weight 16-adapter ensembles, reducing author-specific configuration time from days to minutes. Outperformed grid-search baselines by 23% in privacy-utility Pareto efficiency. - Productionized the agentic workflow on HPC clusters utilizing Mistral-24B for robust evaluation and policy refinement. Engineered multi-objective feedback loops to enable real-time policy iteration, maintaining 88% semantic similarity and 5% perplexity drift.	
Applied Scientist 2 JPMorgan Chase & Co.	Jun 2023 - Aug 2025 Hyderabad, TS
- Led fraud detection initiatives saving over \$220M annually by developing high-recall XGBoost and CatBoost classifiers; optimized recall-precision tradeoffs using constrained objective tuning while maintaining feature interpretability for stakeholders. - Productionized Semi-Supervised Graph Neural Networks across 60M+ transaction nodes; introduced topology-aware edge sampling and dynamic neighborhood aggregation, boosting fraud-detection precision by 9.2% over 2M transactions/day. - Architected a GPU-served contrastive multimodal Transformer (ViT + BERT-style decoder), scaling to 500K+ requests/day, for OCR-based check fraud detection. Improved fraud precision by 19% while reducing latency via batched mixed-precision inference. - Built a Learning-to-Rank information retrieval system over 3B+ KYC records using Pairwise Ranking loss to flag fraud entities; Outperformed the existing TF-IDF baselines, cutting 62% false positives and saving 700+ analyst review hours daily in production. - Developed highly-scalable ML platforms and distributed ETL pipelines in AWS Glue for terabyte-scale financial data. Developed automated feature selection workflows that boosted downstream fraud capture by 14%, reducing manual feature engineering overhead.	
Research Intern McGill University - <i>Mitacs Scholar</i>	Jun 2022 - Sept 2022 Montreal, QC
Developed RoBERTa-based sentence embeddings on AAC image captions to capture word-context. Achieved Krippendorff's alpha reliability score of 0.82 for context-aware vocab retrieval; deployed the model in ClickAAC mobile application.	
Research Intern Samsung Research	Jan 2022 - Jun 2022 Bangalore, KN
Optimized information retrieval precision for Samsung Bixby via bipartite graphs, reducing false-positives by 12%. Drove high-perf INT8 inference by implementing quantization-aware training, matching FP32 perf for word-disambiguation tasks with a 2.9% F1 gain.	

PROJECTS

Agentic Ad Intelligence Engine [\[code\]](#) Architected a **multi-agent orchestration** pipeline for X (Twitter) using **Grok 4.1** to generate 100+ dynamic user personas and personalized ad copy, achieving **sub-5-second** end-to-end latency. Engineered an LLM-driven routing system utilizing real-time telemetry and feedback-driven ranking for dynamic ad delivery; drove a **16% CTR** lift across pilot campaigns.

KeyCognition: Behavioral AI Platform [\[code\]](#) Engineered a temporal sequence modeling pipeline in Databricks, processing **136M+** **real-time keystroke** events to construct latent behavioral embeddings for **cognitive load prediction**. Developed a hybrid behavioral classification model fusing LSTMs and K-Means clustering to extract micro-temporal typing anomalies like backspaces, flight times etc.

PUBLICATIONS

- [Srivastava, H.](#) (2023). Multi-Modal Sentiment Analysis using Text and Audio for Customer Support Centers
- [Srivastava, H.](#) (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems
- [Srivastava, H.](#) (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications

SKILLS

CS/Tooling	C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, CUDA, LangGraph, HuggingFace
AI/Research	Machine Learning, Natural Language Processing, Generative AI, Large Language Models (LLMs), Graph-ML, Ranking, Agentic Workflows, Time Series Forecasting
Infra/MLOps	AWS, Kubernetes, LangChain, MLFlow, RAG, Distributed Systems, Feature Stores, Vector Search